

## 2.1 Measure Preliminaries-1

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\* Let  $X$  be a non empty set. An algebra of sets of  $X$  is a non-empty collection  $\mathcal{A}$  of subsets of  $X$  closed under finite union and complement.

A  $\sigma$ -algebra on  $X$  is a collection  $\mathcal{M}$  of subsets of  $X$  satisfying

- (ci)  $\emptyset \in \mathcal{M}$
- (ii) If  $E \in \mathcal{M}$  then  $E^c = X \setminus E \in \mathcal{M}$ .
- (ciii) If  $\{E_n\}_{n \in \mathbb{N}}$  is a seq<sup>n</sup> of sets in  $\mathcal{M}$ , then  $\bigcup_{n=1}^{\infty} E_n \in \mathcal{M}$ .

Now  $(X, \mathcal{M})$  is a measurable space and elements of  $\mathcal{M}$  are called measurable set.

Properties of  $\sigma$ -algebra.

- 1)  $X \in \mathcal{M}$
- 2) If  $E_1, \dots, E_N \in \mathcal{M}$  then  $\bigcup_{i=1}^N E_i \in \mathcal{M}$
- 3) If  $E_1, \dots, E_N \in \mathcal{M}$  then  $\bigcap_{i=1}^N E_i \in \mathcal{M}$
- 4) If  $\{E_i\}_{i \in \mathbb{N}}$  is a seq<sup>n</sup> of sets in  $\mathcal{M}$ , then  $\bigcap_{i=1}^{\infty} E_i \in \mathcal{M}$
- 5) If  $E, F \in \mathcal{M}$  then  $E \setminus F \in \mathcal{M}$ .

Proof:

- 1)  $\emptyset \in \mathcal{M}$  and  $X = (\emptyset)^c \in \mathcal{M}$ .
- 2) Let  $\{E_i\}_{i \in \mathbb{N}}$  be a countable collection of measurable sets where  $E_i = \emptyset$  for  $i > N$ . Hence,  $\bigcup_{i=1}^{\infty} E_i = E_1 \cup \dots \cup E_N \in \mathcal{M}$ .
- 4)  $(\bigcup_{i=1}^{\infty} E_i)^c = \bigcap_{i=1}^{\infty} E_i^c \in \mathcal{M}$
- 3) Apply 4) to collection of sets in 2)
- 5)  $E \setminus F = E \cap F^c \in \mathcal{M}$  as finite intersection

is closed under  $\sigma$ -algebra.

**\*\*Proposition\*\***

If  $\{M_\alpha \mid \alpha \in J\}$  is collection of  $\sigma$ -algebra on  $X$ . Then  $\bigcap_{\alpha \in J} M_\alpha$  is a  $\sigma$ -algebra.

**\*\*Proof:\*\***

Observe  $\emptyset \in M_\alpha$  for all  $\alpha \in J$  this implies  $\emptyset \in \bigcap_{\alpha \in J} M_\alpha$ . If  $E \in \bigcap_{\alpha \in J} M_\alpha$  then  $E \in M_\alpha$  for all  $\alpha \in J$  hence  $E^c \in M_\alpha$  for all  $\alpha \in J$  this implies  $E^c \in \bigcap_{\alpha \in J} M_\alpha$ . If  $\{E_n\}_{n \in \mathbb{N}}$  is a seq<sup>n</sup> of sets in  $\bigcap_{\alpha \in J} M_\alpha$  hence  $\bigcup_{n \in \mathbb{N}} E_n \in M_\alpha$  for each  $\alpha \in J$ . Hence  $\bigcup_{n \in \mathbb{N}} E_n \in \bigcap_{\alpha \in J} M_\alpha$ .

□

\* In particular let  $S \subseteq \mathcal{P}(X)$ , there exists a unique smallest  $\sigma$ -algebra on  $X$  that contains  $S$ . This is called  $\sigma$ -algebra generated by  $S$ .

*Remark 0.1.* We show uniqueness by constructing a collection of  $\sigma$ -algebra containing  $S$  and then

taking an intersection over the family of  $\sigma$ -algebras. By the above proposition, the resulting collection is the smallest  $\sigma$ -algebra containing  $S$ .

\* Let  $X$  be a topological space. The  $\sigma$ -algebra generated by topology on  $X$  is called Borel  $\sigma$ -algebra on  $X$  and denoted by  $B_X$ . The members of the  $\sigma$ -algebra are called Borel sets.

Note:  $F_\sigma$  is countable union of closed sets and  $G_\delta$  is countable intersection of open sets. These are examples of Borel sets.

Proposition

$\mathcal{B}_{\mathbb{R}}$  is generated by each where  $\mathcal{B}_{\mathbb{R}}$  is borel  $\sigma$ -algebra.

- a. open intervals  $\mathcal{E}_1$
- b. closed intervals  $\mathcal{E}_2$
- c. half-open intervals  $\mathcal{E}_3$
- d. open rays  $\mathcal{E}_4$
- e. closed rays  $\mathcal{E}_5$

Proof: Observe  $\mathcal{E}_1, \mathcal{E}_2, \mathcal{E}_4$  and  $\mathcal{E}_5$  are open or closed sets. For elements  $\mathcal{E}_3$ , these are  $G_\delta$  sets i.e.  $(a, b] = \bigcap_{n=1}^{\infty} (a, b + \frac{1}{n})$ . Hence all are borel sets. If  $\mathcal{M}(\mathcal{E}_j)$  is  $\sigma$ -algebra generated by  $\mathcal{E}_j$ . Then  $\mathcal{M}(\mathcal{E}_j) \subseteq \mathcal{B}_{\mathbb{R}}$  for  $j = 1, 2, 3, 4, 5$ .

As any open set in  $\mathbb{R}$  can be expressed as union of countable open intervals. Hence  $B_{\mathbb{R}} \subseteq M(\mathcal{E})$ .

Now observe for any open interval  $(a, b)$

$$\begin{aligned} (a, b) &= \bigcap_{n=1}^{\infty} \left[ a - \frac{1}{n}, b + \frac{1}{n} \right] = \left( a + \frac{1}{n} \right) \cap \left[ a - \frac{1}{n}, b \right) \\ &= (a, \infty) \cap \bigcap_{n=1}^{\infty} \left( -\infty, b + \frac{1}{n} \right) \\ &= \bigcap_{n=1}^{\infty} \left[ a - \frac{1}{n}, \infty \right) \cap (-\infty, b) \end{aligned}$$

Hence  $M(\mathcal{E}_j) = B_{\mathbb{R}}$  for  $j = 1, 2, 3, 4, 5$ .

**\*\*Bonus!\*\*** Every open set in  $\mathbb{R}$  is a countable union of open intervals.

Let  $U$  be open set in  $\mathbb{R}$ . If  $x \in U$  then  $x$  is rational or irrational. Let  $x$  be rational then construct  $I_x = \bigcup_{x \in I} I$  where  $I$  are non disjoint open intervals containing  $x$  and contained in  $U$ . If  $x$  is not a rational for some  $\epsilon > 0$ ,  $x \in (a - \epsilon, a + \epsilon) \subseteq U$  choose a rational  $q \in (a - \epsilon, a + \epsilon)$  and observe

$x \in I_q$ . Hence for  $x \in U$  there exists  $q \in \mathbb{Q}$  st.  $x \in I_q$ . This implies  $U \subseteq \bigcup_{q \in \mathbb{Q}} I_q$ . Observe  $I_q \subseteq U$  for all  $q \in \mathbb{Q}$  hence

$$U = \bigcup_{q \in \mathbb{Q}} I_q \text{ where } I_q \text{ are open intervals.}$$

\* A family of set  $\mathcal{R} \subseteq \mathcal{P}(X)$  is called a ring if it closed under finite union and difference (i.e. if  $E_1, \dots, E_n \in \mathcal{R}$  then  $\bigcup_{i=1}^n E_i \in \mathcal{R}$  and if  $E, F \in \mathcal{R}$  then  $E \setminus F \in \mathcal{R}$ ). A ring that is closed under countable unions is called a  $\sigma$ -ring.

**\*\*Proposition\*\***

If  $\mathcal{R}$  is a  $\sigma$ -ring are closed under countable intersection.

**\*\*Proof:\*\*** For  $F \in \mathcal{R}$  observe  $\emptyset = F \setminus F \in \mathcal{R}$ . Let  $\{E_i\}_{i=1}^\infty$  be a collection of sets belonging to  $\mathcal{R}$ . If any of them are pairwise disjoint  $\bigwedge_{i=1}^n E_i = \emptyset \in \mathcal{R}$ . Assume  $\{E_i\}_{i=1}^\infty$  are not pairwise disjoint. As  $\mathcal{R}$  is a  $\sigma$ -ring, let  $\bigcup_{i=1}^n E_i = E \in \mathcal{R}$ . Let  $E'_i = E \setminus E_i \in \mathcal{R}$  for  $i \in \mathbb{N}$ . Now  $E' = \bigcup_{i=1}^\infty E'_i \in \mathcal{R}$ . Observe  $\bigwedge_{i=1}^n E_i = E \setminus E' \in \mathcal{R}$ .

\* Let  $X$  be a set equipped with  $\sigma$ -algebra  $\mathfrak{M}$

We define countably additive measure  $m$  such that

- (i)  $m : \mathfrak{M} \rightarrow [0, \infty]$  is well defined function
- (ii)  $m(\emptyset) = 0$ .
- (iii) If  $\{E_j\}_{j=1}^\infty$  is a seq<sup>n</sup> of disjoint sets then

$$m\left(\bigcup_{j=1}^\infty E_j\right) = \sum_{j=1}^\infty m(E_j)$$

Observe (iii) implies finite additivity by taking  $E_j = \emptyset$  for  $j > n$ .

\* We define  $(X, \mathfrak{M}, m)$  as a measure space.

If  $m(X) < +\infty$  we call  $m$  as finite measure

If  $X = \bigcup_{j=1}^\infty E_j$  where  $E_j \in \mathfrak{M}$  and  $m(E_j) < +\infty$  for all  $j$  then  $m$  is  $\sigma$ -finite measure.

More generally  $E$  is  $\sigma$ -finite set if  $E = \bigcup_{j=1}^\infty E_j$

$E_j \in \mathfrak{M}$  and  $m(E_j) < +\infty$  for all  $j$ .

\* If for each  $E \in \mathcal{M}$  with  $m(E) = \infty$  and there exists  $F \subseteq E$  st.  $0 < m(F) < \infty$  then  $m$  is semifinite.

**\*\*Exercise\*\***

Every  $\sigma$ -finite measure is semifinite.

**\*\*Proof:\*\***

If  $m(E) = \infty$  then observe  $E = E \cap X = \bigcup_{j=1}^\infty (E \cap E_j)$  this implies  $E \cap E_j \subseteq E_j \subseteq X$  for some  $j$ . We know  $m(E_j) < +\infty$  hence  $m(E \cap E_j) < m(E_j) < +\infty$ . Hence we have found a subset of  $E$  specifically  $E \cap E_j$  which has measure of  $E_j \cap E$  to be finite.

*Example 0.1.* (i) Let  $X$  be any set and  $x_0 \in X$  be a fixed point and  $M = \mathcal{P}(X)$ . Let  $m : M \rightarrow \mathbb{R}$

$$m(E) = \begin{cases} 1 & \text{if } x_0 \in E \\ 0 & \text{if } x_0 \notin E \end{cases}$$

This is called the Dirac measure.

(ii) Let  $X$  be any set and  $M = \mathcal{P}(X)$ . Let  $m : M \rightarrow [0, \infty]$

$$m(E) = \begin{cases} |E| & \text{if } |E| < +\infty \\ \infty & \text{otherwise} \end{cases}$$

where  $|\cdot|$  is cardinality function and  $m$  is called the counting measure.

(iii) Let  $X$  be any set and  $M = \mathcal{P}(X)$ . Let  $m : M \rightarrow [0, \infty]$

$$m(E) = +\infty \text{ for all } E \subset \mathbb{R}$$

**Lemma 0.1.** Suppose  $X$  be a set and  $\mathfrak{M}$  is a  $\sigma$ -algebra equipped on  $X$ . Let  $m : \mathfrak{M} \rightarrow [0, \infty]$  be a measure, then

(i) [Monotonicity] : If  $A, B \in \mathfrak{M}$  s.t.  $A \subseteq B$  then

$$m(A) \leq m(B).$$

(ii) If  $\{A_n\} \subseteq \mathfrak{M}$  then

$$m\left(\bigcup_{n \in \mathbb{N}} A_n\right) \leq \sum_{n \in \mathbb{N}} m(A_n).$$

*Proof.* (i) If  $B \supseteq A$  then  $m(B) = m(A \cup B \setminus A)$

$$= m(A) + m(B \setminus A) \geq m(A).$$

(ii) Let  $E_1 = A_1$ , and  $E_k = A_k \setminus \bigcup_{i=1}^{k-1} A_i$  for  $k \geq 1$ . Then observe all  $E_k$  are disjoint to each other hence

$$m\left(\bigcup_{k \in \mathbb{N}} E_k\right) = \sum_{k \in \mathbb{N}} m(E_k)$$

then note

$$\bigcup_{n \in \mathbb{N}} A_n = \bigcup_{k \in \mathbb{N}} E_k$$

and  $m(E_k) \leq m(A_k)$  for all  $k \in \mathbb{N}$ . Hence we get

$$m\left(\bigcup_{n \in \mathbb{N}} A_n\right) \leq \sum_{k \in \mathbb{N}} m(A_k).$$

□

**Lemma 0.2.** Let  $(X, \mathcal{M}, \mu)$  be a measure space.

(a) Continuity from below: If  $\exists E_j \uparrow_j^\infty = 1 \subset \mathcal{M}$  and  $E_1 \subset E_2 \subset \dots$ , then  $\mu(\bigcup_j E_j) = \lim_{j \rightarrow \infty} \mu(E_j)$ .

(b) Continuity from above: If  $\exists E_j \downarrow_j^\infty = 1 \subset \mathcal{M}$  and  $E_1 \supset E_2 \supset E_3 \supset \dots$ , then  $\mu(\bigcap_j E_j) = \lim_{j \rightarrow \infty} \mu(E_j)$ . and  $\mu(E_1) < +\infty$ .

*Proof.* (a) Set  $E_0 = \emptyset$ . Observe  $\bigcup_j E_j = \bigcup_{j=1}^{\infty} E_j \setminus E_{j-1}$ . Hence,  $\mu(\bigcup_j E_j) = \mu(\bigcup_{j=1}^{\infty} E_j \setminus E_{j-1})$   
 $= \sum_{j=1}^{\infty} \mu(E_j \setminus E_{j-1}) = \lim_{n \rightarrow \infty} \sum_{j=1}^n \mu(E_j \setminus E_{j-1})$   
 $= \lim_{n \rightarrow \infty} \mu(E_n)$ .

(b) Let  $F_j = E_1 \setminus E_j$ . Observe  $F_1 \subset F_2 \subset \dots$  and  $\bigcup_{j=1}^{\infty} F_j = E_1 \setminus \bigcap_j E_j$ .  
 $\mu(E_1) = \mu(\bigcap_j E_j) + \lim_{n \rightarrow \infty} \mu(F_n)$  □

$$\begin{aligned} \mu(E_i) &= \mu\left(\bigcap_{j=1}^{\infty} E_j\right) + \lim_{n \rightarrow \infty} \mu(E_i \setminus E_j) \\ &= \mu\left(\bigcap_{j=1}^{\infty} E_j\right) + \mu(E_i) - \lim_{n \rightarrow \infty} \mu(E_j). \end{aligned}$$

This implies  $\mu(\bigcap_{j=1}^{\infty} E_j) = \lim_{n \rightarrow \infty} \mu(E_j)$ , as  $\mu(E_i) < +\infty$ , we can subtract it from both sides.

□

\* If  $(X, \mathcal{M}, \mu)$  is a measure space, a set  $E \in \mathcal{M}$  such that  $\mu(CE) = 0$  is called a null set.

If a statement is true about point  $x \in X \setminus E$  for some null set  $E$  then we say the statement holds "almost everywhere" (a.e) (More precisely, we have  $\mu$ -Null set and  $\mu$ -a.e).

**\*\*Corollary\*\***

Any countable union of null sets is a null set, and subsets of null sets belonging in  $\mathcal{M}$  are null sets.

\* If a measure whose domain include all subsets of a null set then the domain is called complete.

**Theorem 0.1.** Suppose that  $(X, \mathcal{M}, \mu)$  is a measure space. Let  $\mathcal{N} = \{N \in \mathcal{M} : \mu(N) = 0\}$  and  $\overline{\mathcal{M}} = \{E \cup F \mid E \in \mathcal{M} \text{ and } F \subset N \text{ for some } N \in \mathcal{N}\}$ . Then  $\overline{\mathcal{M}}$  is a  $\sigma$ -algebra and there is a unique extension  $\overline{\mu}$  of  $\mu$  to complete measure  $\overline{\mathcal{M}}$ .

*Proof.* (a)  $\overline{\mathcal{M}}$  is a  $\sigma$ -algebra

Since  $\mathcal{M}$  and  $\mathcal{N}$  are closed under countable unions, so is  $\overline{\mathcal{M}}$ . If  $E \in \mathcal{M}$  and  $F \subset N \in \mathcal{N}$  then  $E \cup F \in \overline{\mathcal{M}}$ . Now assume  $E \cap N = \emptyset$  otherwise replace  $F$  by  $F \setminus E$  and  $N$  by  $N \setminus E$ . Then  $E \cup F = (E \cup N) \cap (N^c \cup F)$ , so  $(E \cup F)^c = (E \cup N)^c \cup (N^c \cup F)^c$  and observe

$(E \cup N)^c \in \mathcal{M}$  and  $N \setminus F^c = N \setminus F \subset N$  and  $N \setminus F \in \mathcal{N}$ . Hence  $(E \cup F)^c \in \overline{\mathcal{M}}$ .

(cb) Define  $\overline{\mu} : \overline{\mathcal{M}} \rightarrow [0, \infty]$ , such that  $\overline{\mu}(E \cup F) = \mu(CE)$ . We want to show well definedness. Let  $E_1 \cup F_1 = E_2 \cup F_2$  where  $E_1, E_2 \in \mathcal{M}$  and  $F_1, F_2 \in \mathcal{N}$ . Now,  $E_1 \subset E_1 \cup F_1 = E_2 \cup F_2 \subset E_2 \cup N_2$ . This implies  $\mu(CE_1) \leq \mu(E_2) + \mu(N_2) = \mu(E_2)$ . If  $E_2 \subset E_1 \cup N_1$ , and this implies  $\mu(E_2) \leq \mu(E_1)$ . Hence  $\mu(CE_1) = \mu(CE_2)$ .

(c) Uniqueness. Let  $m_1, m_2$  be extension of  $\mu$  as per the above extension. Now,  $E \cup F \in \overline{\mathcal{M}}$ . Observe  $m_1(E \cup F) = \mu(CE) = m_2(E \cup F')$  where  $F'$  is any subset of a null set.

\* Outer measure

\* An outer measure on a non-empty set  $X$  is a function  $\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$  that satisfy,

\*  $\mu^*(\emptyset) = 0$ .

\*  $\mu^*(A) \leq \mu^*(B)$  if  $A \subseteq B$ .

\*  $\mu^*(\bigcup_{n=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} \mu^*(A_n)$

**\*\*Proposition\*\***

Let  $E \subseteq \mathcal{P}(X)$  and  $p : E \rightarrow [0, \infty]$  be such that  $\emptyset \in E$ ,  $X \in E$  and  $p(\emptyset) = 0$ . For any  $A \subset X$  we define

$$\mu^*(A) = \inf \left\{ \sum_{i=1}^{\infty} p(E_j) \mid E_j \in E \text{ and } A \subseteq \bigcup_{j=1}^{\infty} E_j \right\}$$

Then  $\mu^*$  is an outer measure.

Proof: (a) Observe a countable collection of empty sets covers  $A = \emptyset$  hence

$$\mu^*\emptyset = \inf \sum_{n \in \mathbb{N}} \rho(E_n) \leq \sum_{n \in \mathbb{N}} \rho(\emptyset) = 0$$

Also  $\rho(E_n) \geq 0$  for any  $E_n \in \mathcal{E}$  hence

$$\sum \rho(E_n) \geq 0$$

for any collection of intervals therefore

$$\inf \sum_{E_n \in \mathcal{E}} \rho(E_n) \geq 0.$$

So we get  $\mu^*\emptyset = 0$ .

For any non-empty set  $A \subset X$  there exists  $\{E_j\}_{j=1}^{\infty} \subset \mathcal{E}$  such that  $A \subseteq \bigcup_{j=1}^{\infty} E_j$  we can set  $E_j = X$  for all  $j$ .

(cb) If  $A \subseteq B$  then if  $\{E_j\}$  is collection of sets covering  $B$  then  $\mathfrak{B}$  covers  $A$  as well.

If  $\mathfrak{B}$  is a cover of  $B$ . We construct

$$A = \{E_n \in \mathfrak{B} \mid B_n \cap A \neq \emptyset\}$$

and observe  $A \subseteq \mathfrak{B}$  Hence

$$\sum_{A_n \in A} P(A_n) \leq \sum_{B_n \in \mathfrak{B}} P(B_n)$$

therefore  $\mu^*A \leq \sum_{B_n \in \mathfrak{B}} P(B_n)$ . Observe  $\mu^*A \leq \mu^*B$  otherwise  $\mu^*A > \mu^*B$  then there exist a countable collection  $\{E_j\}^2 = E$  covering  $B$  and also  $A$  whose sum i.e.  $\sum P(B_n)$  is less than  $\mu^*A$  which contradicts the fact that  $\mu^*A$  is the infimum of sum of countable collection of sets in  $E$  covering  $A$ .

(c) Suppose  $\{A_j\}_{j=1}^{\infty} \subset \mathcal{P}(X)$  and fix  $\delta > 0$ . For each  $j \in \mathbb{N}$  we get  $\{E_{j,k}\}_{k=1}^{\infty} \subset \mathcal{E}$  such that

$$A_j \subset \bigcup_{k=1}^{\infty} E_{j,k} \quad \text{and} \quad \sum_{k=1}^{\infty} \rho(E_{j,k}) \leq \mu^*(A_j) + \frac{\delta}{2^j}.$$

But if  $A = \bigcup_{j=1}^{\infty} A_j$ , we have  $A \subset \bigcup_{j,k=1}^{\infty} E_{j,k}$  and

$$\sum_{j,k} \rho(E_{j,k}) \leq \sum_j \rho(E_{j,k}) + \delta, \quad \text{whence}$$

$$\mu^*(A) \leq \sum_j \mu^*(A_j) + \delta. \quad \text{As } \delta \text{ is arbitrary, we are done.}$$

□

\* A set  $A$  is measurable if for each set  $E$  we have  $m^*E = m^*(A \cap E) + m^*(A^c \cap E)$

Note: We know for any set  $m^*E \leq m^*(A \cap E) + m^*(A^c \cap E)$  holds. Hence  $A$  is measurable iff

$$m^*E \geq m^*(A \cap E) + m^*(A^c \cap E) \quad \text{for any set } E.$$

*Remark 0.2.* If  $A$  is measurable then  $A^c$  is measurable.

**Lemma 0.3.** If  $A_1$  and  $A_2$  are measurable sets then  $A_1 \cup A_2$  is measurable.

*Proof.* For any set  $E \subset X$

$$\mu^*(E \cap (A_1 \cup A_2)) = \mu^*(E \cap A_1) + \mu^*(E \cap A_2 \cap A_1^c)$$

and

$$\mu^*(E \cap (A_1 \cup A_2)) + \mu^*(E \cap (A_1 \cup A_2)^c) = \mu^*(E \cap A_1) + \mu^*(E \cap A_2 \cap A_1^c) + \mu^*(E \cap A_2^c \cap A_1^c)$$

Now apply the fact that  $A_2$  is a measurable set,

$$\mu^*(E \cap (A_1 \cup A_2)) + \mu^*(E \cap (A_1 \cup A_2)^c) = \mu^*(E \cap A_1) + \mu^*(E \cap A_2^c) = \mu^*(E)$$

as  $A_1$  is a measurable set. Hence  $A_1 \cup A_2$  is measurable. □

**Lemma 0.4.** Let  $E$  be any set, and  $A_1, \dots, A_n$  a finite sequence of disjoint measurable sets. Then

$$\mu^* \left( E \cap \left( \bigcup_{i=1}^n A_i \right) \right) = \sum_{i=1}^n \mu^*(E \cap A_i)$$

*Proof.* Base case holds for  $n = 1$ . Assume the given holds for  $n - 1$  sets.

Observe  $E \cap \left( \bigcup_{i=1}^{n-1} A_i \right) \cap A_n = E \cap A_n$  and

$$E \cap \left( \bigcup_{i=1}^{n-1} A_i \right) \cap A_n^c = E \cap \bigcup_{i=1}^{n-1} A_i$$

as  $A_n$  is disjoint from  $\xi_{i=1}^{n-1} A_i$ . Hence

$$\begin{aligned} \mu^* \left( E \cap \left( \bigcup_{i=1}^n A_i \right) \right) &= \\ \mu^* \left( E \cap \bigcup_{i=1}^{n-1} A_i \cap A_n \right) &+ \mu^* \left( E \cap \bigcup_{i=1}^{n-1} A_i \cap A_n^c \right) \end{aligned}$$

by measurability of  $A_n$ . This implies

$$\mu^* \left( E \cap \left( \bigcup_{i=1}^n A_i \right) \right) = \mu^*(E \cap A_n) + \mu^* \left( E \cap \bigcup_{i=1}^{n-1} A_i \right)$$

□

By induction hypothesis's, we get

$$\mu^* \left( E \cap \left( \bigcup_{i=1}^n A_i \right) \right) = \sum_{i=1}^n \mu^*(E \cap A_i).$$

□

**Theorem 0.2.** (Carathéodory's Theorem) If  $\mu^*$  is an outer measure on  $X$ , the collection  $\mathcal{M}$  of  $\mu^*$ -measurable sets is a  $\sigma$ -algebra and the restriction of  $\mu^*$  to  $\mathcal{M}$  is a complete measure.

*Proof.* According to a previous remark and lemma,  $\mathcal{M}$  is an algebra and finitely additive. As per the previous lemma,

$$\mu^* \left( E \cap \left( \bigcup_{i=1}^{\infty} A_i \right) \right) = \sum_{i=1}^{\infty} \mu^*(E \cap A_i).$$

holds for any  $E \subset X$  and collection of  $\mu^*$ -measurable disjoint sets  $\{A_i\}_{i=1}^{\infty}$ . □

In order to  $\mathcal{M}$  is closed under disjoint unions.

Let  $B_n = \bigcup_{j=1}^n A_j$  and  $B = \bigcup_{j=1}^{\infty} A_j$ . Hence

$$\begin{aligned} \mu^*(E) &= \mu^*(E \cap B_n) + \mu^*(E \cap B_n^c) \\ &\geq \sum_{j=1}^n \mu^*(E \cap A_j) + \mu^*(E \cap B^c) \end{aligned}$$

observe the statement above holds for any arbitrary  $n$ .

$$\mu^*(E) \geq \sum_{j=1}^{\infty} \mu^*(E \cap A_j) + \mu^*(E \cap B^c)$$



$$\begin{aligned}
&\geq \mu^* \left( \bigcup_{j=1}^{\infty} (E \cap A_j) \right) + \mu^*(E \cap B^c) \\
&= \mu^*(E \cap B) + \mu^*(E \cap B^c) \geq \mu^*(E).
\end{aligned}$$

It follows that  $B \in \mathcal{M}$  and  $\mu^*(B) = \sum_{j=1}^{\infty} \mu^*(A_j)$ , so  $\mu^*$  is countably additive, by substituting  $E = B$ .

Finally, if  $\mu^*(A) < \infty$  for any  $E \subset X$  we have

$$\begin{aligned}
\mu^*(E) &\leq \mu^*(E \cap A) + \mu^*(E \cap A^c) \\
&= \mu^*(E \cap A^c) \leq \mu(E).
\end{aligned}$$

so,  $A \in \mathcal{M}$ . Hence  $\mu^*|_{\mathcal{M}}$  is complete measure

\* Let  $A \subset \mathcal{P}(X)$  is an algebra, a function  $\mu_0 : A \rightarrow [0, \infty]$  will be called a premeasure if  $\mu_0(\emptyset) = 0$ ,  
\* if  $\{A_j\}_{j \in \mathbb{N}}$  is a sequence of disjoint sets in  $A$ , then  $\mu_0(\bigcup_{j=1}^{\infty} A_j) = \sum_{j=1}^{\infty} \mu_0(A_j)$ .

**\*\*Note:\*\*** The difference between a measure and a premeasure is the domain.

\* Measure is defined on a  $\sigma$ -algebra \* Premeasure is defined on an algebra.

\* We have a notion of finite and  $\sigma$ -finite premeasures as same as that of a measure.

*Remark 0.3.* If  $\mu_0$  is a premeasure on  $\mathcal{A} \subseteq \mathcal{PCX}$ , it induces an outer measure on  $X$  in accordance with Proposition /ref , namely

$$\mu^*(E) = \inf \left\{ \sum_{j=1}^{\infty} \mu_0(A_j) \mid A_j \in \mathcal{A}, E \subseteq \bigcup_{j=1}^{\infty} A_j \right\}$$

**Lemma 0.5.** If  $\mu_0$  is a premeasure on  $\mathcal{A}$  and  $\mu^*$  as defined in remark above, then

1.  $\mu^*|_{\mathcal{A}} = \mu_0$ ;
2. every set in  $\mathcal{A}$  is  $\mu^*$ -measurable.

*Proof.* a) Suppose  $E \in \mathcal{A}$ . If  $E \subseteq \bigcup_{j=1}^{\infty} A_j$  with any countable collection  $\{A_j\}_{j=1}^{\infty} \subseteq \mathcal{A}$  then we let  $B_n = E \cap \left( A_n \setminus \bigcup_{j=1}^{n-1} A_j \right)$ . Observe  $B_n$  is a disjoint member of  $\mathcal{A}$  whose union is  $E$ , so  $\square$

$\mu_0(E) = \sum_{n=1}^{\infty} \mu_0(B_n) \leq \sum_{n=1}^{\infty} \mu_0(A_j)$ . Since this holds for any arbitrary cover of  $E$  hence as per the definition of  $\mu^*$ , we get  $\mu_0(E) \leq \mu^*(E)$ . Now as  $E \in \mathcal{A}$ , we can choose  $A_j \subset \bigcup_{j=1}^{\infty} A_j$  st.  $A_1 = E$  and  $A_j = \emptyset$ . Hence  $\mu_0(E) = \mu^*(E)$ .

b) If  $A \in \mathcal{A}$ ,  $E \subset X$  and  $\epsilon > 0$  then there is a sequence  $B_j \subset_{j \in \mathbb{N}} A$  with  $E \subset \bigcup_{j=1}^{\infty} B_j$  and  $\sum_{j=1}^{\infty} \mu_0(B_j) \leq \mu^*(E) + \epsilon$ . Since  $\mu_0$  is additive on  $\mathcal{A}$ ,

$$\mu_0(\bigcup_{j=1}^{\infty} B_j \cap X) = \mu_0(\bigcup_{j=1}^{\infty} B_j \cap (A \cup A^c))$$

$$\begin{aligned}
&= \mu_0(\bigcup_{j=1}^{\infty} (B_j \cap A)) + \mu_0(\bigcup_{j=1}^{\infty} (B_j \cap A^c)) \\
&= \sum_{j=1}^{\infty} \mu_0(B_j \cap A) + \sum_{j=1}^{\infty} \mu_0(B_j \cap A^c).
\end{aligned}$$

Hence,

$$\begin{aligned}
\mu^*(E) + \epsilon &\geq \sum_{j=1}^{\infty} \mu_0(B_j \cap A) + \sum_{j=1}^{\infty} \mu_0(B_j \cap A^c) \\
&\geq \mu_0(E \cap A) + \mu_0(E \cap A^c)
\end{aligned}$$

As  $\epsilon$  is arbitrary,  $A$  is  $\mu^*$ -measurable.

**Theorem 0.3.** Let  $A \subseteq \mathcal{P}(X)$  be an algebra,  $\mu_0$  is a premeasure defined on  $A$ . Let  $\mathcal{M}$  be  $\sigma$ -algebra generated by  $A$ . There exists a measure  $\mu$  on  $\mathcal{M}$  such that  $\mu|_A = \mu_0$ .

*Proof.* If we define  $\mu^*$  as defined earlier and  $\mu := \mu^*|_{\mathcal{M}}$ . Then we know by the previous lemma  $\mu^*|_A = \mu|_A = \mu_0$  and all  $A$  are  $\mu^*$ -measurable. This implies  $\sigma$ -algebra of  $\mu^*$ -measurable sets include  $A$ .  $\square$

**Theorem 0.4.** If  $\nu$  is another measure defined on  $\mathcal{M}$  (as per the previous theorem) that extends on  $\mu_0$  then  $\nu(E) \leq \mu(E)$  for any  $E \in \mathcal{M}$  and equality when  $\mu(E) < +\infty$ . If  $\mu_0$  is  $\sigma$ -finite, then  $\mu$  is the unique extension of  $\mu_0$  to a measure on  $\mathcal{M}$ .

*Proof.* If  $E \in \mathcal{M}$  and  $E \subset \bigcup_{j=1}^{\infty} A_j$  where  $A_j \in \mathcal{A}$  for all  $j \in \mathbb{N}$ , then  $\nu(CE) \leq \sum_{j=1}^{\infty} \nu(A_j) = \sum_{j=1}^{\infty} \mu_0(A_j)$  holds true for any  $\{A_j\}_{j=1}^{\infty}$  collection. Hence,

$$\nu(E) \leq \mu(CE).$$

Let  $A = \bigcup_{j=1}^{\infty} A_j$ . Now  $\nu(A) = \nu\left(\bigcup_{j=1}^{\infty} A_j\right)$

$$\begin{aligned}
&= \sum_{j=1}^{\infty} \nu(A_j) = \lim_{n \rightarrow \infty} \sum_{j=1}^n \nu(A_j) = \lim_{n \rightarrow \infty} \nu\left(\bigcup_{j=1}^n A_j\right) \\
&= \lim_{n \rightarrow \infty} \mu_0\left(\bigcup_{j=1}^n A_j\right) = \sum_{j=1}^{\infty} \mu_0(A_j) = \mu_0(A).
\end{aligned}$$

$\square$

If  $\mu(E) < +\infty$ . For every  $\epsilon > 0$  there exist a collection of sets  $\{A_{\epsilon, j}\}_{j=1}^{\infty}$ , such that  $A_{\epsilon} = \bigcup_{j=1}^{\infty} A_{\epsilon, j}$  such that  $\mu(A_{\epsilon}) < \mu(CE) + \epsilon$ , hence  $\mu(A_{\epsilon} \setminus E) < \epsilon$ , and  $\mu(CE) \leq \mu(A) = \nu(A) = \nu(E) + \nu(A \setminus E) \leq \nu(CE) + \mu(A \setminus E) \leq \nu(CE) + \epsilon$  holds for all  $\epsilon > 0$ . This implies  $\mu(CE) \leq \nu(CE)$ .

Finally suppose  $X = \bigcup_j A_j$  with  $\mu_0(A_j) < \infty$  where we can assume  $A_j$ 's are disjoint (or reconstruct a given collection into that way). Then for any  $E \in \mathcal{M}$

$$\begin{aligned}
\mu(E) &= \sum_j \mu(E \cap A_j) = \sum_j \nu(CE \cap A_j) \\
&= \nu(CE),
\end{aligned}$$

so  $\mu = \nu$ .

A: finite disjoint unions of h-intervals is an algebra.

Borel Measure on  $\mathbb{R}$

Notation:  $(a, b]$  are denoted as h-intervals.

$\mu$  is the borel measure on  $\mathcal{B}_{\mathbb{R}}$ . We define distribution function as  $F(x) = \mu((-\infty, x])$

Proposition

Let  $F : \mathbb{R} \rightarrow \mathbb{R}$  be an increasing and right cont. If  $\{(a_j, b_j]\}_{j=1}^n$  are disjoint h-intervals, let

$$\mu_0 \left( \bigcup_{j=1}^n (a_j, b_j] \right) = \sum_{j=1}^n \mu_0((a_j, b_j]),$$

and let  $\mu_0(\emptyset) = 0$ . Then  $\mu_0$  is a premeasure on algebra  $A$ .

Proof: (a)  $\mu_0$  is well defined

Given any collection of disjoint h-intervals. If  $\{(a_j, b_j]\}_{j=1}^n$  by relabelling, we get

$$a = a_1 < b_1 = a_2 < \dots < b_n = b.$$

This implies  $\sum_{j=1}^n [F(b_j) - F(a_j)] = F(b) - F(a)$ .

More generally, if  $\{I_i\}_{i=1}^n$  and  $\{J_j\}_{j=1}^m$  are finite sequences of disjoint  $h$ -intervals such that  $\bigcup_{i=1}^n I_i = \bigcup_{j=1}^m J_j$  by the reasoning above we can say,

$$\sum_i \mu_0(I_i) = \sum_{i,j} \mu_0(I_i \cap J_j) = \sum_j \mu_0(J_j).$$

(b)  $\mu_0$  is countably additive

Observe  $\mu_0$  is finitely additive by construction.

We want to show if  $\sum I_i \subseteq \mathcal{U}$  st.  $\bigcup_{i \in \mathbb{N}} I_i \in \mathcal{U}$  then

$$\mu_0 \left( \bigcup_{i=1}^{\infty} I_i \right) = \sum_{i=1}^{\infty} \mu_0(I_i).$$

As  $\bigcup_{i=1}^{\infty} I_i \in \mathcal{U}$ , we can separate  $\{I_i\}_{i=1}^{\infty}$  into finitely many subsequences whose union is an  $h$ -interval.

Assume  $\bigcup_{i=1}^{\infty} I_i$  is an  $h$ -interval (otherwise you look into a smaller collection of  $I_i$ 's whose union is an  $h$ -interval.)

$I = \bigcup_{i=1}^{\infty} I_i = [a, b)$ . Observe

$$\mu_0(I) = \mu_0 \left( \bigcup_{j=1}^n I_j \right) + \mu_0 \left( I \setminus \bigcup_{j=1}^n I_j \right)$$

$$\geq \mu_0 \left( \bigcup_{j=1}^n I_j \right) = \sum_{j=1}^n \mu_0(I_j)$$

holds for any  $n \in \mathbb{N}$ , we obtain  $\mu_0(I) \geq \sum_{j=1}^n \mu(I_j)$ .

To show reverse inequality

case (ci): If  $a, b$  are finite.

As  $F$  is right continuous, for any  $\epsilon > 0$ , there exists  $\delta_\epsilon > 0$  such that  $F(a + \delta_\epsilon) - F(a) < \epsilon$ .

If  $I_j = (a_j, b_j]$   $j \in \mathbb{N}$  then we can find a  $\delta_{\epsilon,j} > 0$  such that  $F(b_j + \delta_{\epsilon,j}) - F(b_j) < \frac{\epsilon}{2^j}$ .

Now  $\{(a_j, b_j + \delta_{\epsilon,j})\}_{j=1}^\infty$  is a collection of open intervals covering  $[a + \delta_\epsilon, b]$ , so there is a finite subcover.

Let  $(a_1, b_1 + \delta_1), \dots, (a_N, b_N + \delta_N)$  cover  $[a + \delta_\epsilon, b]$  and  $b_i + \delta_i \in (a_{i+1}, b_{i+1} + \delta_{i+1})$ .

(We have replaced  $\delta_{\epsilon,i} = \delta_i$  for all  $i = \{1, \dots, N-1\}$ ).

For any  $\epsilon > 0$ , observe

$$\begin{aligned} \mu_0(I) &< F(b) - F(a + \delta_\epsilon) + \epsilon \\ &\leq F(b_N + \delta_N) - F(a_1) + \epsilon \\ &= F(b_N + \delta_N) - F(a_N) + \sum_{i=1}^{N-1} [F(a_{i+1}) - F(a_i)] + \epsilon \\ &\leq F(b_N + \delta_N) - F(a_N) + \sum_{i=1}^N [F(b_j + \delta_j) - F(a_j)] + \epsilon \\ &< \left( \sum_{i=1}^N F(b_j) + \frac{\epsilon}{2^j} - F(a_j) \right) + \epsilon < \sum_{i=1}^\infty \mu(I_j) + 2\epsilon. \end{aligned}$$

As the above statement holds for all  $\epsilon > 0$ , we are done.

**\*\*case (ii):\*\*** If one of them is NOT finite.

Say  $a = -\infty$ . Fix  $\epsilon > 0$ .

Then for any  $M > 0$ , we can find an  $t$  open cover  $(a_j + b_j + \delta_j)$  covering  $[-M, b]$ . Hence by the same reasoning we get

$$F(b) - F(-M) \leq \sum_{j=1}^\infty \mu_0(I_j) + 2\epsilon,$$

whereas if  $b = \infty$ , for any  $M < \infty$  we get

$$F(M) - F(a) \leq \sum_{j=1}^{\infty} \mu_0(I_j) + 2\varepsilon.$$

We get the desired result by  $\varepsilon \rightarrow 0$  followed by  $M \rightarrow \infty$ . (Go through this).

**Theorem 0.5.** If  $F : \mathbb{R} \rightarrow \mathbb{R}$  is any increasing, right continuous  $d^m$ , there is a unique Borel measure  $\mu_F$  on  $\mathbb{R}$  such that  $\mu_F((a, b]) = F(b) - F(a)$  for all  $a, b$ . If  $G$  is another function similar to this then  $G - F$  is constant iff  $\mu_F = \mu_G$

Conversely if  $\mu$  is a Borel measure on  $\mathbb{R}$  that is finite on all bounded Borel sets and we define

$$F(x) = \begin{cases} \mu((0, x]) & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -\mu((x, 0]) & \text{if } x < 0 \end{cases}$$

*Proof.* Each  $F$  induces a premeasure on  $\mathcal{U}$  by previous proposition. It is clear that  $F$  and  $G$  induce the same premeasure iff  $F - G$  is constant, and these premeasure are  $\sigma$ -finite as  $\mathbb{R} = \bigcup_{j \in \mathbb{Z}} (j, j+1]$ .  $\square$

The first two assertions therefore follow from Theorem. As for the converse, the monotonicity of  $\mu$  implies  $F$  to be an increasing function.

Observe  $\mu$  is continuous from above and below implies the right continuity of  $F$  for  $x \geq 0$  and  $x < 0$ . It is evident that  $\mu = \mu_F$  on  $\mathcal{U}$ , and hence  $\mu = \mu_F$  on  $\mathcal{B}_{\mathbb{R}}$  by uniqueness of Theorem.

*Remark 0.4.*  $\circ$  The theory could have also been developed for  $[a, b)$  intervals and  $F$  as left cont.  $S^n$ .

$\circ$  If  $\mu$  is a finite Borel measure on  $\mathbb{R}$  then  $F(x) = \mu((-\infty, x])$  is the cumulative distribution function.

$\circ$  Any increasing right cont.  $F^n$  gives us a complete measure.

Given an increasing, right continuous function  $F$ . We get  $(\mathbb{R}, \mathcal{M}_{\mu}, \mu)$  as a complete measure space where for any  $E \in \mathcal{M}_{\mu}$ ,

$$\mu(E) = \inf \left\{ \sum_{j=1}^{\infty} [F(b_j) - F(a_j)] \mid E \subset \bigcup_{j=1}^{\infty} (a_j, b_j] \right\}$$

**Lemma 0.6.** For any  $E \in \mathcal{M}_{\mu}$ ,

$$\mu(E) = \inf \left\{ \sum_{j=1}^{\infty} \mu((a_j, b_j)) \mid E \subset \bigcup_{j=1}^{\infty} (a_j, b_j] \right\}$$

*Proof.* Let  $\nu : \mathcal{M}_{\mu} \rightarrow [0, \infty)$  such that

$$\nu(E) = \inf \left\{ \sum_{j=1}^{\infty} \mu((a_j, b_j)) \mid E \subset \bigcup_{j=1}^{\infty} (a_j, b_j] \right\}$$

and  $\mu$  be the measure defined in above given def<sup>n</sup>.

Goal :  $\nu(E) = \mu(E)$  for all  $E \in \mathcal{M}_\mu$ . □

(i) We want to show  $\mu(E) \leq \nu(E)$  for any  $E \in \mathcal{M}_\mu$ .

Suppose  $E \subseteq \bigcup_{j=1}^{\infty} (a_j, b_j)$  and for each  $(a_j, b_j)$  we get  $\{I_j^k\}_{k=1}^{\infty}$  as disjoint  $h$ -intervals covering  $(a_j, b_j)$  for each  $k$ . Let  $I_j^k = [c_j^k, c_j^{k+1}]$  where  $c_j^1 = a_j$  and  $c_j^k \rightarrow b_j$  as  $k \rightarrow \infty$ . Thus  $E \subseteq \bigcup_{j,k} I_{j,k}$ . Hence

$$\sum_{j=1}^{\infty} \mu((a_j, b_j)) = \sum_{j=1}^{\infty} \mu(I_j^k) \geq \mu(E)$$

(Or you can use  $E \subseteq \bigcup_{j \in \mathbb{N}} (a_j, b_j)$  and by monotonicity and subadditivity and apply infimum, we get  $\mu(E) \leq \nu(E)$ ).

(ii) WTS :  $\nu(E) \leq \mu(E)$ ; for any  $\varepsilon > 0$ , we get  $\{(a_j, b_j)\}_{j=1}^{\infty}$  covering  $E$  and

$$\sum_{j=1}^{\infty} \mu((a_j, b_j)) \leq \mu(E) + \varepsilon.$$

As  $F$  is right cont. and increasing, for  $\epsilon > 0$  we find  $\delta_{\epsilon j} = \delta_j$  for each  $j \in \mathbb{N}$  st.

$$F(b_j + \delta_j) - F(b_j) < \epsilon/2^j.$$

Then  $E \subset \bigcup_{j=1}^{\infty} (a_j + b_j + \delta_j)$  and

$$\begin{aligned} \mu((a_j, b_j + \delta_j)) &\leq \mu((a_j, b_j + \delta_j)) \\ &= F(b_j + \delta_j) - F(a_j) \\ &= F(b_j + \delta_j) - F(b_j) + F(b_j) - F(a_j) \end{aligned}$$

for all  $j \in \mathbb{N}$ . Now by taking a sum

$$\begin{aligned} \sum_{j=1}^{\infty} \mu((a_j, b_j)) &\leq \sum_{j=1}^{\infty} F(b_j + \delta_j) - F(b_j) \\ &\quad + \sum_{j=1}^{\infty} \mu((a_j, b_j)) \\ &\leq \epsilon + \epsilon + \mu(E). \end{aligned}$$

Observe  $\nu(E) \leq \sum_{j=1}^{\infty} \mu((a_j, b_j)) \leq \mu(E) + 2\epsilon$  holds for all  $\epsilon > 0$ . Hence  $\nu(E) \leq \mu(E)$ .

**Theorem 0.6.** If  $E \in \mathcal{M}_\mu$ , then

$$\begin{aligned} \mu(E) &= \inf \{ \mu(U) \mid U \supseteq E \text{ and } U \text{ is open} \} \\ &= \sup \{ \mu(K) \mid K \subseteq E \text{ and } K \text{ is compact} \} \end{aligned}$$